

# 5

## Energy Expenditure, Fatigue, and Muscle Soreness

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**ACTIVITY 5.3** *Sources of Fatigue Affecting Athletic Performance* considers various sources of fatigue and how they affect athletic performance.

## Critical Power: The Link Between Energy Expenditure and Fatigue

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**ACTIVITY 5.4** *Muscle Soreness and Cramps* investigates the causes and treatment of muscle soreness and muscle cramps.

## In Closing

**T**he causes and sites of what exercisers and athletes call “fatigue” are as numerous as the sensations that characterize it. Although it is usually considered in terms of how different parts of the body feel—burning lungs, aching legs, unyielding tiredness—researchers have begun to focus on the role the brain plays in fatigue. After all, the brain collects all of the sensory feedback from the body and determines when physical exertion simply cannot continue. Recent research has shown that a fatigued brain can squash successful sport performance as much as tired muscles can. An article by Dr. Samuele Marcora titled “Mental Fatigue Impairs Physical Performance in Humans,” published in the *Journal of Applied Physiology*, suggests that perceptions of fatigue cause us to reach our physical limits long before the body does. A group of rugby players exercised to fatigue during an endurance test but were subsequently able to do a 5 s sprint. That is, the brain’s perceptions of fatigue stopped the endurance trial before the athletes had reached their physical limits. The brain may be acting as a regulatory brake, slowing down activity before a somatic limit is reached. This does not mean that fatigue is imagined. Rather, its causes are complex (for example, both the muscles and the brain rely on glucose and glycogen for fuel).

One cannot understand exercise physiology without understanding some key concepts about energy expenditure at rest and during exercise. In [chapter 2](#), we discussed the formation of adenosine triphosphate (ATP), the major form of chemical energy stored within, and used by, cells. Adenosine triphosphate is produced from substrates by a series of processes that are known collectively as *metabolism*. In the first half of this chapter we discuss various techniques for measuring the whole-body energy expenditure or metabolic rate, then we describe how energy expenditure varies from basal or resting conditions up to maximal exercise intensities. If exercise is sustained for a prolonged time, eventually muscular contraction cannot be sustained and performance will diminish. This inability to maintain muscle contractions is broadly called fatigue. Fatigue is a complex, multidimensional phenomenon that may or may not result from an inability to maintain metabolism and expend energy. Because fatigue often has a metabolic component, it is discussed in this chapter along with energy expenditure. Muscle

soreness and cramping are also discussed as additional factors that can limit exercise.

## Measuring Energy Expenditure

The energy used by contracting muscle fibers during exercise cannot be directly measured. But numerous laboratory methods can be used to calculate whole-body energy expenditure at rest and during exercise. Several of these methods have been in use since the early 1900s. Others are new and have only recently been used in exercise physiology.

### Direct Calorimetry

Only about 40% of the energy liberated during the metabolism of glucose and fats is used to produce ATP. The remaining 60% is converted to heat, so one way to gauge the rate and quantity of energy production is to measure the body's heat production. This technique is called **direct calorimetry** ("measuring heat"), since the basic unit of heat is the **calorie (cal)**.

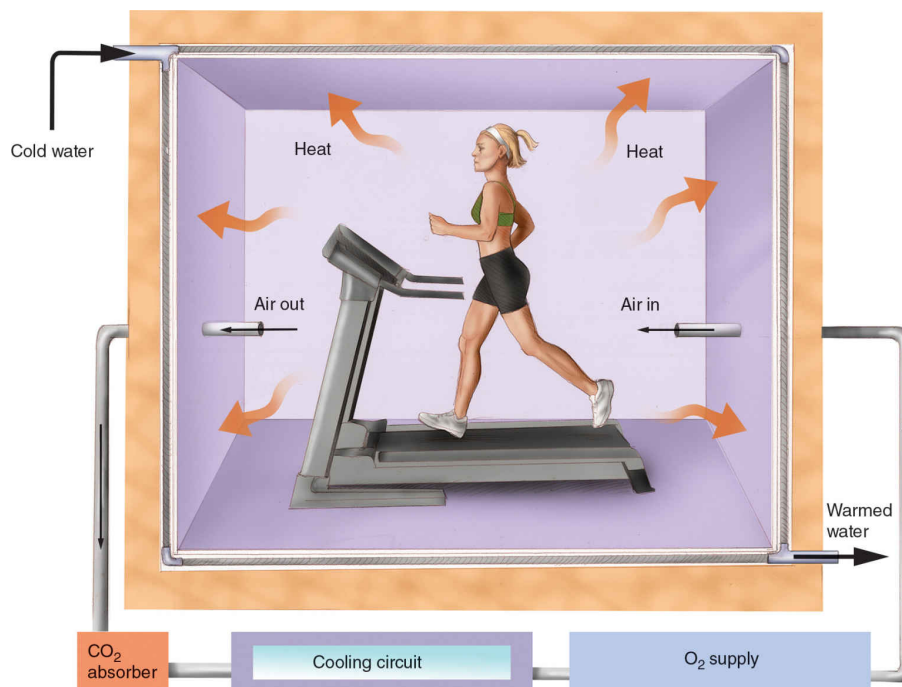
This approach was first described by Zuntz and Hagemann in the late 1800s.<sup>29</sup> They developed the **calorimeter**, which consists of an insulated, airtight chamber as illustrated in [figure 5.1](#). The walls of the chamber contain tubing through which water is circulated. In the chamber, the heat produced by the body radiates to the walls and warms the water. The water temperature and temperature changes of the air entering and leaving the chamber vary with the heat the body generates. One's metabolism can be calculated from the resulting values.

Calorimeters are expensive to construct and operate and are slow to generate results, so very few are in actual operation. Their only real advantage is that they measure heat directly, but they have several disadvantages for exercise physiology. Although a calorimeter can provide an accurate measure of total body energy expenditure over time, it cannot follow rapid changes in energy expenditure. Therefore, while direct calorimetry is useful for measuring resting metabolism and energy expended during prolonged, steady-state aerobic exercise, energy metabolism during more typical exercise situations cannot be adequately studied with a

direct calorimeter. Second, exercise equipment such as a motor-driven treadmill gives off its own heat that must be accounted for in the calculations. Third, not all heat is liberated from the body; some is stored in the body, causing body temperature to rise. And finally, sweating affects the measurements and the constants used in the calculations of heat produced. Consequently, it is easier and less expensive to quantify energy expenditure by measuring the exchange of oxygen and carbon dioxide that occurs during oxidative phosphorylation.

### Indirect Calorimetry

As discussed in [chapter 2](#), oxidative metabolism of glucose and fat—the main substrates for aerobic exercise—uses  $O_2$  and produces  $CO_2$  and water. The rate of  $O_2$  and  $CO_2$  exchanged in the lungs normally equals the rate of their usage and release by the body tissues. Based on this principle, energy expenditure can be determined by measuring the respiratory exchange of  $O_2$  and  $CO_2$ . This method of estimating total body energy expenditure is called **indirect calorimetry** because heat production is not measured directly.



**FIGURE 5.1** A direct calorimeter for the measurement of energy expenditure by an exercising human subject. The heat generated by the subject's body is transferred to the air and walls of the chamber (through conduction, convection, and evaporation). This heat produced by the subject—a measure of his or her metabolic rate—is measured by recording the temperature change in the air entering and leaving the calorimeter as well as in the water flowing through its walls.

In order for oxygen consumption to reflect energy metabolism accurately, energy production must be almost completely oxidative. If a large portion of energy is being produced anaerobically, respiratory gas measurements will not reflect all metabolic processes and will underestimate the total energy expenditure. Therefore, this technique is limited to steady-state aerobic activities lasting a few minutes or longer, which fortunately takes into account most daily activities including exercise.

Respiratory gas exchange is determined through measurement of the volume of  $O_2$  and  $CO_2$  that enters and leaves the lungs during a given period of time. Because  $O_2$  is removed from the inspired air in the alveoli and  $CO_2$  is added to the alveolar air, the expired  $O_2$  concentration is less than the inspired, whereas the  $CO_2$  concentration is higher in expired air than in inspired air. Consequently, the differences in the concentrations of these gases between the inspired and the expired air tell us how much  $O_2$  is being taken up and how much  $CO_2$  is being produced by the body. Because the body has only limited  $O_2$  storage, the amount taken up at the lungs accurately reflects the body's use of  $O_2$ . Although a number of sophisticated and expensive methods are available for measuring the respiratory exchange of  $O_2$  and  $CO_2$ , the simplest and oldest methods (i.e., Douglas bag to collect expired air and chemical analysis of collected gas sample) are probably the most accurate, but they are relatively slow and permit only a few measurements during each session. Modern electronic computer systems for respiratory gas exchange measurements offer the ability to make rapid and repeated measurements.

Notice in [figure 5.2](#) that the gas expired by the subject passes through a hose into a mixing chamber. The subject is wearing a nose clip so that all expired gas is collected from the mouth and none is lost to the air. From the mixing chamber, samples are pumped to electronic oxygen and carbon dioxide analyzers. In this setup, a

computer uses the measurements of expired gas (air) volume and the fraction (percentage) of oxygen and carbon dioxide in a sample of that expired air to calculate  $O_2$  uptake and  $CO_2$  production. Sophisticated equipment can do these calculations breath by breath, but calculations are more typically done over discrete time periods lasting from one to several minutes.

### ***Calculating Oxygen Consumption and Carbon Dioxide Production***

Using equipment like that shown in [figure 5.2](#), exercise physiologists can measure the three variables needed to calculate the actual volume of oxygen consumed ( $VO_2$ ) and volume of  $CO_2$  produced ( $VCO_2$ ). Generally, values are presented as oxygen consumed per minute ( $\dot{V}O_2$ ) and  $CO_2$  produced per minute ( $\dot{V}CO_2$ ). The dot over the V ( $\dot{V}$ ) indicates the rate of  $O_2$  consumption or  $CO_2$  production per minute.

In simplified form,  $\dot{V}O_2$  is equal to the volume of  $O_2$  inspired minus the volume of  $O_2$  expired. To calculate the volume of  $O_2$  inspired, we multiply the volume of air inspired by the fraction of that air that is composed of  $O_2$ ; the volume of  $O_2$  expired is equal to the volume of air expired multiplied by the fraction of the expired air that is composed of  $O_2$ . The same holds true for  $CO_2$ .



**FIGURE 5.2** Typical equipment that is routinely used by exercise physiologists to measure  $O_2$  consumption and  $CO_2$  production. These values can be used to calculate  $\dot{V}O_{2\max}$  and the respiratory exchange ratio and therefore energy expenditure. Although this equipment is cumbersome and limits movement, smaller versions have recently been adapted for use under a variety of conditions in the laboratory, on the playing field, in industry, and elsewhere.

Thus, calculation of  $\dot{V}O_2$  and  $\dot{V}CO_2$  requires the following information:

- Volume of air inspired ( $\dot{V}_I$ )
- Volume of air expired ( $\dot{V}_E$ )
- Fraction of oxygen in the inspired air ( $F_I O_2$ )
- Fraction of  $CO_2$  in the inspired air ( $F_I CO_2$ )
- Fraction of oxygen in the expired air ( $F_E O_2$ )
- Fraction of  $CO_2$  in the expired air ( $F_E CO_2$ )

The oxygen consumption, in liters of oxygen consumed per minute, can then be calculated as follows:

$$\dot{V}O_2 = (\dot{V}_I \times F_I O_2) - (\dot{V}_E \times F_E O_2)$$

The CO<sub>2</sub> production is similarly calculated as follows:

$$\dot{V}\text{CO}_2 = (\dot{V}_E \times F_{E\text{CO}_2}) - (\dot{V}_I \times F_{I\text{CO}_2})$$

These equations provide reasonably good estimates of  $\dot{V}\text{O}_2$  and  $\dot{V}\text{CO}_2$ . However, the equations are based on the idea that inspired air volume exactly equals expired air volume and there are no changes in gases stored within the body. Since there are differences in gas storage during exercise (discussed next), more accurate equations can be derived from the variables listed.

### ***Haldane Transformation***

Over the years, scientists have attempted to simplify the actual calculation of oxygen consumption and CO<sub>2</sub> production. Several of the measurements needed in the preceding equations are known and do not change. The gas concentrations of the three gases that make up inspired air are known: oxygen accounts for 20.93% (or 0.2093), CO<sub>2</sub> accounts for 0.03% (0.0003), and nitrogen accounts for 79.03% (0.7903) of the inspired air. What about the volume of inspired and expired air? Aren't they the same, such that we would need to measure only one of the two?

Inspired air volume equals expired air volume only when the volume of O<sub>2</sub> consumed equals the volume of CO<sub>2</sub> produced. When the volume of oxygen consumed is greater than the volume of CO<sub>2</sub> produced,  $\dot{V}_I$  is greater than  $\dot{V}_E$ . Likewise,  $\dot{V}_E$  is greater than  $\dot{V}_I$  when the volume of CO<sub>2</sub> produced is greater than the volume of oxygen consumed. However, the one thing that is constant is that the volume of nitrogen inspired ( $\dot{V}_I \text{ N}_2$ ) is equal to the volume of nitrogen expired ( $\dot{V}_E \text{ N}_2$ ). Because  $\dot{V}_I \text{ N}_2 = \dot{V}_I \times F_{I \text{ N}_2}$  and  $\dot{V}_E \text{ N}_2 = \dot{V}_E \times F_{E \text{ N}_2}$ , we can calculate  $\dot{V}_I$  from  $\dot{V}_E$  by using the following equation, which has been referred to as the **Haldane transformation**:

$$(1) \dot{V}_I \times F_{I \text{ N}_2} = \dot{V}_E \times F_{E \text{ N}_2},$$

which can be rewritten as

$$(2) \dot{V}_I = (\dot{V}_E \times F_{E \text{ N}_2}) / F_{I \text{ N}_2}.$$

Furthermore, because we are actually measuring the concentrations of O<sub>2</sub> and CO<sub>2</sub> in the expired gases, we can calculate  $F_{E \text{ N}_2}$  from the

sum of  $F_{EO_2}$  and  $F_{ECO_2}$ , or

$$(3) F_{EN_2} = 1 - (F_{EO_2} + F_{ECO_2}).$$

So, in pulling all of this information together, we can rewrite the equation for calculating  $\dot{V}O_2$  as follows:

$$\dot{V}O_2 = (\dot{V}_I \times F_{IO_2}) - (\dot{V}_E \times F_{EO_2})$$

By substituting equation 2, we get the following:

$$\dot{V}O_2 = [(\dot{V}_E \times F_{EN_2}) / (F_{IN_2} \times F_{IO_2})] - [(\dot{V}_E) \times (F_{EO_2})]$$

By substituting known values for  $F_{IO_2}$  of 0.2093 and for  $F_{IN_2}$  of 0.7903, we get the following:

$$\dot{V}O_2 = [(\dot{V}_E \times F_{EN_2}) / (0.7903 \times 0.2093)] - (\dot{V}_E \times F_{EO_2})$$

By substituting equation 3, we get the following:

$$\dot{V}O_2 = \{(\dot{V}_E) \times [1 - (F_{EO_2} + F_{ECO_2})] \times (0.2093 / 0.7903)\} - (\dot{V}_E \times F_{EO_2})$$

or, simplified,

$$\dot{V}O_2 = (\dot{V}_E) \times \{[1 - (F_{EO_2} + F_{ECO_2})] \times 0.265\} - (\dot{V}_E \times F_{EO_2})$$

or, further simplified,

$$\dot{V}O_2 = (\dot{V}_E) \times \{[1 - (F_{EO_2} + F_{ECO_2})] \times 0.265\} - (F_{EO_2}).$$

This final equation is the one actually used in practice by exercise physiologists, although computers now do the calculating automatically in most laboratories.

One final correction is necessary. When air is expired, it is at body temperature (BT), is at the prevailing atmospheric or ambient pressure (P), and is saturated (S) with water vapor, or what are referred to as BTPS conditions. Each of these influences would not only add error to the measurement of  $\dot{V}O_2$  and  $\dot{V}CO_2$  but also would make it difficult to compare measurements made in laboratories at different altitudes, for example. For that reason, every gas volume is routinely converted to its standard temperature (ST: 0 °C or 273 K) and pressure (P: 760 mmHg), dry equivalent (D), or STPD. This is accomplished by a series of correction equations.

## ***Respiratory Exchange Ratio***

To estimate the amount of energy used by the body, it is necessary to know the type of food substrate (combination of carbohydrate, fat, protein) being oxidized. The carbon and oxygen contents of glucose, free fatty acids (FFAs), and amino acids differ dramatically. As a result, the amount of oxygen used during metabolism depends on the type of fuel being oxidized. Indirect calorimetry measures the rate of CO<sub>2</sub> release ( $\dot{V}\text{CO}_2$ ) and oxygen consumption ( $\dot{V}\text{O}_2$ ). The ratio between these two values is termed the **respiratory exchange ratio (RER)**.

$$\text{RER} = \dot{V}\text{CO}_2 / \dot{V}\text{O}_2$$

In general, the amount of oxygen needed to completely oxidize a molecule of carbohydrate or fat is proportional to the amount of carbon in that fuel. For example, glucose (C<sub>6</sub>H<sub>12</sub>O<sub>6</sub>) contains six carbon atoms. During glucose combustion, six molecules of oxygen are used to produce 6 CO<sub>2</sub> molecules, 6 H<sub>2</sub>O molecules, and 32 ATP molecules:



By evaluating how much CO<sub>2</sub> is released compared with the amount of O<sub>2</sub> consumed, we find that the RER is 1.0:

$$\text{RER} = \dot{V}\text{CO}_2 / \dot{V}\text{O}_2 = 6 \text{ CO}_2 / 6 \text{ O}_2 = 1.0$$

As shown later in the chapter, the RER value varies with the type of fuels being used for energy. Free fatty acids have considerably more carbon and hydrogen but less oxygen than glucose. Consider palmitic acid, C<sub>16</sub>H<sub>32</sub>O<sub>2</sub>. To completely oxidize this molecule to CO<sub>2</sub> and H<sub>2</sub>O requires 23 molecules of oxygen:

$$\begin{array}{r} 16 \text{ C} + 16 \text{ O}_2 \rightarrow 16 \text{ CO}_2 \\ 32 \text{ H} + 8 \text{ O}_2 \rightarrow 16 \text{ H}_2\text{O} \\ \hline \text{Total} = \quad 24 \text{ O}_2 \text{ needed} \\ \quad \quad -1 \text{ O}_2 \text{ provided by the palmitic acid} \\ \hline \quad \quad 23 \text{ O}_2 \text{ must be added} \end{array}$$

Ultimately, this oxidation results in 16 molecules of CO<sub>2</sub>, 16 molecules of H<sub>2</sub>O, and 129 molecules of ATP:



Combustion of this fat molecule requires significantly more oxygen than combustion of a carbohydrate molecule. During carbohydrate oxidation, approximately 6.3 molecules of ATP are produced for each molecule of  $\text{O}_2$  used (32 ATP per 6  $\text{O}_2$ ), compared with 5.6 molecules of ATP per molecule of  $\text{O}_2$  during palmitic acid metabolism (129 ATP per 23  $\text{O}_2$ ).

Although fat provides more energy than carbohydrate, more oxygen is needed to oxidize fat than carbohydrate. This means that the RER value for fat is substantially lower than for carbohydrate. For palmitic acid, the RER value is 0.70:

$$\text{RER} = \dot{V}\text{CO}_2 / \dot{V}\text{O}_2 = 16 / 23 = 0.70$$

Once the RER value is determined from the calculated respiratory gas volumes, the value can be compared with a table ([table 5.1](#)) to determine the food mixture being oxidized. If, for example, the RER value is 1.0, the cells are using only glucose or glycogen, and each liter of oxygen consumed would generate 5.05 kcal. The oxidation of only fat would yield 4.69 kcal/L of  $\text{O}_2$ , and the oxidation of protein would yield 4.46 kcal/L of  $\text{O}_2$  consumed. Thus, if the muscles were using only glucose and the body were consuming 2 L of  $\text{O}_2$ /min, then the rate of heat energy production would be 10.1 kcal/min (2 L/min · 5.05 kcal/L).

### ***Limitations of Indirect Calorimetry***

While indirect calorimetry is a common and extremely important tool of exercise physiologists, it has some limitations. Calculations of gas exchange assume that the body's  $\text{O}_2$  content remains constant and that  $\text{CO}_2$  exchange in the lung is proportional to its release from the cells. Arterial blood remains almost completely oxygen saturated (about 98%), even during intense effort, and we can accurately assume that the oxygen being removed from the air we breathe is in proportion to its cellular uptake. Carbon dioxide exchange, however, is less constant. Body  $\text{CO}_2$  pools are quite large and can be altered simply by deep breathing or by performance of highly intense exercise. Under these conditions, the amount of  $\text{CO}_2$  released in the lung may not represent that being produced in the tissues, so

calculations of carbohydrate and fat used based on gas measurements are accurate only at rest or during steady-state exercise.

Use of the RER can also lead to inaccuracies. Recall that protein is not completely oxidized in the body. This makes it impossible to calculate the body's protein use from the RER. As a result, the RER is sometimes referred to as nonprotein RER because it simply ignores any protein oxidation. Traditionally, protein was thought to contribute little to the energy used during exercise, so exercise physiologists felt justified in using the nonprotein RER when making calculations. But more recent evidence suggests that in exercise lasting for several hours, protein may contribute up to 5% of the total energy expended under certain circumstances.

**TABLE 5.1 Respiratory Exchange Ratio (RER) as a Function of Energy Derived From Various Fuel Mixtures**

| % Kcal from   |      |      |                                 |
|---------------|------|------|---------------------------------|
| Carbohydrates | Fats | RER  | Energy (kcal/L O <sub>2</sub> ) |
| 0             | 100  | 0.71 | 4.69                            |
| 16            | 84   | 0.75 | 4.74                            |
| 33            | 67   | 0.80 | 4.80                            |
| 51            | 49   | 0.85 | 4.86                            |
| 68            | 32   | 0.90 | 4.92                            |
| 84            | 16   | 0.95 | 4.99                            |
| 100           | 0    | 1.00 | 5.05                            |

The body normally uses a combination of fuels. Respiratory exchange ratio values vary depending on the specific mixture being oxidized. At rest, the RER value is typically in the range of 0.78 to 0.80. During exercise, though, muscles rely increasingly on carbohydrate for energy, resulting in a higher RER. As exercise intensity increases, the muscles' carbohydrate demand also increases. As more carbohydrate is used, the RER value approaches 1.0.

This increase in the RER value to 1.0 reflects the demands on blood glucose and muscle glycogen, but it also may indicate that more CO<sub>2</sub> is being unloaded from the blood than is being produced by the muscles. At or near exhaustion, lactate accumulates in the blood. The body tries to reverse this acidification by releasing more CO<sub>2</sub>. Lactate accumulation increases CO<sub>2</sub> production because

excess acid causes carbonic acid in the blood to be converted to  $\text{CO}_2$ . As a consequence, the excess  $\text{CO}_2$  diffuses out of the blood and into the lungs for exhalation, increasing the amount of  $\text{CO}_2$  released. For this reason, RER values approaching 1.0 may not accurately estimate the type of fuel being used by the muscles.

Another complication is that glucose production from the catabolism of amino acids and fats in the liver produces an RER below 0.70. Thus, calculations of carbohydrate oxidation from the RER value will be underestimated if energy is derived from this process.

Despite its shortcomings, indirect calorimetry still provides the best estimate of energy expenditure at rest and during aerobic exercise and is widely used in laboratories throughout the world.

### **Isotopic Measurements of Energy Metabolism**

In the past, determining an individual's total daily energy expenditure depended on recording food intake over several days and measuring body composition changes during that period. This method, although widely used, is limited by the individual's ability to keep accurate records and by the ability to match the individual's activities to accurate energy costs.

Fortunately, the use of chemical isotopes has expanded our ability to investigate energy metabolism. Isotopes are elements with an atypical atomic weight. They can be either radioactive (radioisotopes) or nonradioactive (stable isotopes). As an example, carbon-12 ( $^{12}\text{C}$ ) has a molecular weight of 12, is the most common natural form of carbon, and is nonradioactive. In contrast, carbon-14 ( $^{14}\text{C}$ ) has two more neutrons than  $^{12}\text{C}$ , giving it an atomic weight of 14.  $^{14}\text{C}$  is radioactive.

Carbon-13 ( $^{13}\text{C}$ ) constitutes about 1% of the carbon in nature and is used frequently for studying energy metabolism. Because  $^{13}\text{C}$  is nonradioactive, it is less easily traced within the body than  $^{14}\text{C}$ . But although radioactive isotopes are easily detected in the body, they pose a hazard to body tissues and thus are used infrequently in human research.

$^{13}\text{C}$  and other isotopes such as hydrogen-2 (deuterium, or  $^2\text{H}$ ) are used as tracers, meaning that they can be selectively followed in the

body. Tracer techniques involve infusing isotopes into an individual and then following their distribution and movement.

Although the method was first described in the 1940s, studies that used doubly labeled water for monitoring energy expenditure during normal daily living in humans were not conducted until the 1980s. The subject ingests a known amount of water labeled with two isotopes ( $^2\text{H}_2$  and  $^{18}\text{O}$ ), hence the term *doubly labeled water*. The deuterium ( $^2\text{H}$ ) diffuses throughout the body's water, and the oxygen-18 ( $^{18}\text{O}$ ) diffuses throughout both the water and the bicarbonate stores (where much of the  $\text{CO}_2$  derived from metabolism is stored). The rate at which the two isotopes leave the body can be determined by analysis of their presence in a series of urine, saliva, or blood samples. These turnover rates then can be used to calculate how much  $\text{CO}_2$  is produced, and that value can be converted to energy expenditure through the use of calorimetric equations.

Because isotope turnover is relatively slow, energy metabolism must be measured for several weeks. Thus, this method is not well suited for measurements of acute exercise metabolism. However, its accuracy (more than 98%) and low risk make it well suited for determining day-to-day energy expenditure.

## In Review

- Direct calorimetry involves a large sophisticated chamber that directly measures heat produced by the body; while it can provide very accurate measures of resting metabolism, it is not a commonly used tool for exercise physiologists.
- Indirect calorimetry involves measuring whole-body  $\text{O}_2$  consumption and  $\text{CO}_2$  production from expired gases. Since we know the fraction of  $\text{O}_2$  and  $\text{CO}_2$  in the inspired air, three additional measurements are needed: the volume of air inspired ( $\dot{V}_I$ ) or expired ( $\dot{V}_E$ ), the fraction of oxygen in the expired air ( $F_{E\text{O}_2}$ ), and the fraction of  $\text{CO}_2$  in the expired air ( $F_{E\text{CO}_2}$ ).
- By calculating the RER (the ratio of  $\text{CO}_2$  production to  $\text{O}_2$  consumption) and determining the metabolic substrates being oxidized, we can convert  $\dot{V}\text{O}_2$  into energy expenditure in kilocalories.
- The RER value at rest is usually 0.78 to 0.80. The RER value for the oxidation of fat is 0.70 and is 1.00 for carbohydrates.

- Isotopes can be used to determine metabolic rate over longer periods of time. They are injected or ingested into the body. The rates at which they are cleared can be used to calculate CO<sub>2</sub> production and then caloric expenditure.
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## Energy Expenditure at Rest and During Exercise

With the techniques described in the previous section, exercise physiologists can measure the amount of energy a person expends in a variety of conditions. This section deals with the body's rates of energy expenditure (metabolic rates) at rest, during submaximal and maximal exercise, and during the period of recovery following an acute exercise bout.

### Basal and Resting Metabolic Rates

The rate at which the body uses energy is called the metabolic rate. Estimates of energy expenditure at rest and during exercise are often based on measurement of whole-body oxygen consumption ( $\dot{V}O_2$ ) and its caloric equivalent. At rest, an average person consumes about 0.3 L of O<sub>2</sub>/min.

Knowing an individual's  $\dot{V}O_2$  allows us to calculate that person's caloric expenditure. Recall that at rest, the body usually burns a mixture of carbohydrate and fat. An RER value of approximately 0.80 is fairly common for most resting individuals eating a mixed diet. The caloric equivalent associated with an RER value of 0.80 is 4.80 kcal per liter of O<sub>2</sub> consumed (see [table 5.1](#)). Using these values and an estimate of 0.3 L of O<sub>2</sub>/min, we can calculate this individual's caloric expenditure as follows:

$$\begin{aligned}\text{kcal/day} &= \text{liters of O}_2 \text{ consumed per day} \times \text{kcal used per liter of O}_2 \\ &= 432 \text{ L O}_2/\text{day} \times 4.80 \text{ kcal/L O}_2 \\ &= 2,074 \text{ kcal/day}\end{aligned}$$

This value closely agrees with the average resting energy expenditure expected for a 70 kg (154 lb) man. Of course, it does not include the extra energy needed for normal daily activity or any excess energy used for exercise.

One standardized measure of energy expenditure at rest is the **basal metabolic rate (BMR)**. The BMR is the rate of energy

expenditure for an individual at rest in a supine position, measured in a thermoneutral environment immediately after at least 8 h of sleep and at least 12 h of fasting. This value reflects the minimum amount of energy required to carry on essential physiological functions.

Because muscle has high metabolic activity, the BMR is directly related to an individual's fat-free mass and is generally reported in kilocalories per kilogram of fat-free mass per minute ( $\text{kcal} \cdot \text{kg FFM}^{-1} \cdot \text{min}^{-1}$ ). The higher the fat-free mass, the more total calories expended in a day. Because women tend to have a lower fat-free mass and a greater percent body fat than men, women tend to have a lower BMR than men of a similar weight.

Body surface area also affects BMR. The higher the surface area, the more heat loss occurs from the skin, which raises the BMR because more energy is needed to maintain body temperature. For this reason, the BMR is sometimes reported in kcal per square meter of body surface area per hour ( $\text{kcal} \cdot \text{m}^{-2} \cdot \text{h}^{-1}$ ). Because we are discussing daily energy expenditure, we will use the simpler unit, kcal/day.

Many other factors affect BMR, including these:

- Age: BMR gradually decreases with increasing age, generally because of a decrease in fat-free mass.
- Body temperature: BMR increases with increasing temperature.
- Psychological stress: Stress increases activity of the sympathetic nervous system, which increases the BMR.
- Hormones: For example, increased release of thyroxine from the thyroid gland or epinephrine from the adrenal medulla can both increase the BMR.

Instead of BMR, most researchers measure **resting metabolic rate (RMR)**, which is similar to BMR but does not require the stringent standardized conditions associated with a true BMR. Basal metabolic rate and RMR values are typically within 5% to 10% of each other, with BMR slightly lower, and range from 1,200 to 2,400 kcal/day. But the average total metabolic rate of an individual engaged in normal daily activity ranges from 1,800 to 3,000 kcal. However, the energy expenditure for large athletes engaged in

intense training—for example, large football players in two-a-day practice sessions—can exceed 10,000 kcal/day!

### Metabolic Rate During Submaximal Exercise

Exercise increases the energy requirement well in excess of RMR. Metabolism increases in direct proportion to the increase in exercise intensity, as shown in [figure 5.3a](#). As this subject exercised on a cycle ergometer for 5 min at 50 watts (W), oxygen consumption ( $\dot{V}O_2$ ) increased from its resting value to a steady-state value within 1 min or so. The same subject then cycled for 5 min at 100 W, and again a steady-state  $\dot{V}O_2$  was reached in 1 to 2 min. In a similar manner, the subject cycled for 5 min at 150 W, 200 W, 250 W, and 300 W, respectively, and steady-state values were achieved at each power output. The steady-state  $\dot{V}O_2$  value represents the energy cost for that specific power output. The steady-state  $\dot{V}O_2$  values were plotted against their respective power outputs (right half in [figure 5.3a](#)), showing clearly that there is a linear increase in the  $\dot{V}O_2$  with increases in power output.

From more recent studies, it is clear that the  $\dot{V}O_2$  response at higher rates of work does not follow the steady-state response pattern shown in [figure 5.3a](#) but rather looks more like the graphs presented in [figure 5.3b](#). At power outputs above the lactate threshold (the lactate response is indicated by the dashed line in the right half of [figure 5.3](#), *a* and *b*), the oxygen consumption continues to increase beyond the typical 1 to 2 min needed to reach a steady-state value. This increase has been called the slow component of oxygen uptake kinetics.<sup>11</sup> The most likely mechanism for this slow component is an alteration in muscle fiber recruitment patterns, with the recruitment of more type II muscle fibers, which are less efficient (i.e., they require a higher  $\dot{V}O_2$  to achieve the same power output).<sup>11</sup>

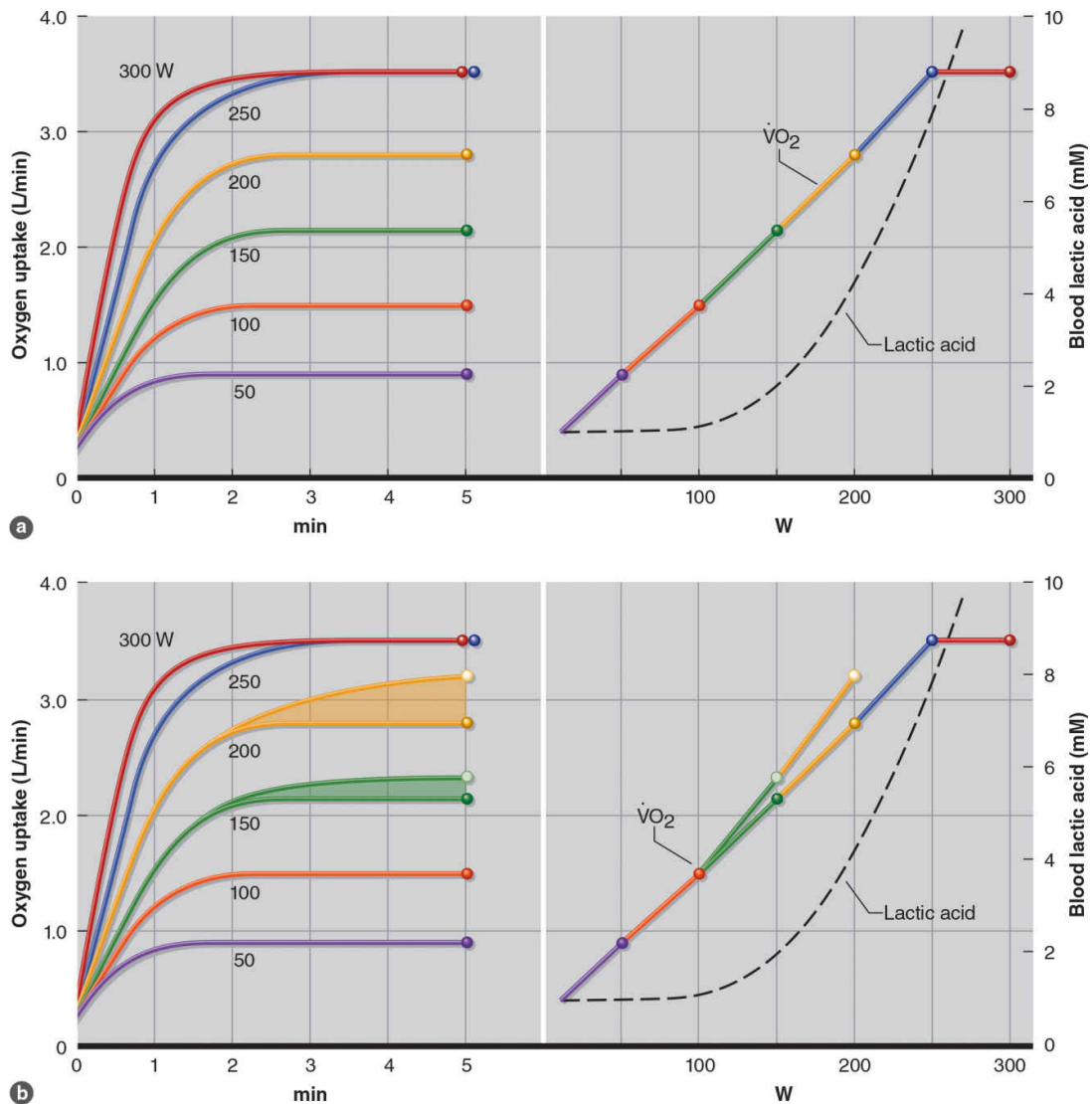
A similar, but unrelated, phenomenon is referred to as the  **$\dot{V}O_2$  drift**, defined as a slow increase in  $\dot{V}O_2$  during prolonged, submaximal, constant power output exercise. Unlike the slow component,  $\dot{V}O_2$  drift is observed at power outputs well below lactate threshold, and the increase in  $\dot{V}O_2$  drift is more gradual. Although not understood completely,  $\dot{V}O_2$  drift is likely attributable to

an increase in ventilation and effects of increased circulating catecholamines.

### Maximal Capacity for Aerobic Exercise

In [figure 5.3a](#), it is clear that when the subject cycled at 300 W, the  $\dot{V}O_2$  response was not different from that achieved at 250 W. This indicates that the subject had reached the maximal limit of his ability to increase his  $\dot{V}O_2$ . This value is referred to as aerobic capacity, **maximal oxygen uptake ( $\dot{V}O_{2max}$ )**.  $\dot{V}O_{2max}$  is widely regarded as the best single measurement of **cardiorespiratory endurance** or aerobic fitness. This concept is further illustrated in [figure 5.4](#), which compares the  $\dot{V}O_{2max}$  of a trained and an untrained man.

In some exercise settings, as intensity increases, a subject reaches volitional fatigue before a plateau occurs in the  $\dot{V}O_2$  response (the criterion for a true  $\dot{V}O_{2max}$ ). In such cases, the highest oxygen uptake achieved is more correctly termed the **peak oxygen uptake ( $\dot{V}O_{2peak}$ )**. For example, a highly trained marathon runner will almost always achieve a higher  $\dot{V}O_2$  value ( $\dot{V}O_{2max}$ ) on a treadmill than when he or she is tested to volitional fatigue on a cycle ergometer ( $\dot{V}O_{2peak}$ ). In the latter case, fatigue of the quadriceps muscles is likely to occur before a true maximal oxygen uptake is achieved.



**FIGURE 5.3** The increase in oxygen uptake with increasing power output (a) as originally proposed by P.-O. Åstrand and K. Rodahl, *Textbook of work physiology: Physiological bases of exercise*, 3rd ed. (New York: McGraw-Hill, 1986), p. 300; and (b) as redrawn by Gaesser and Poole (1996, p. 36). See the text for a detailed explanation of this figure.

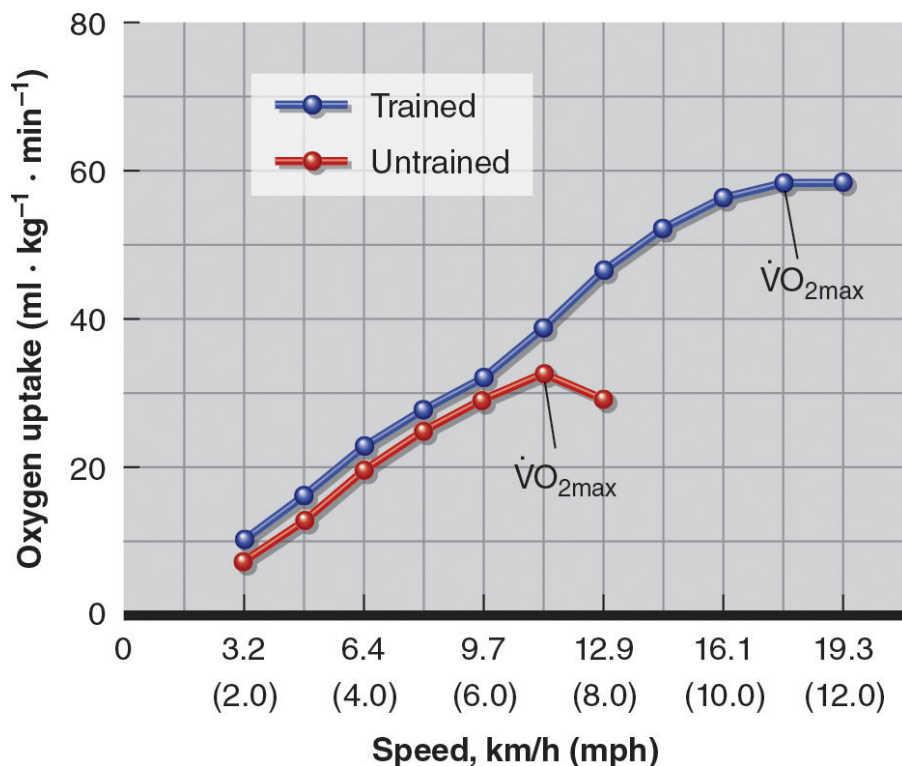
Reprinted by permission from G.A. Gaesser and D.C. Poole, "The Slow Component of Oxygen Uptake Kinetics in Humans," *Exercise and Sport Sciences Reviews* 24 (1996): 36.



Although  $\dot{V}O_{2\max}$  is a good measure of aerobic fitness, the winner of a marathon race cannot be predicted from the runner's laboratory-

measured  $\dot{V}O_{2\max}$ . This suggests that while a relatively high  $\dot{V}O_{2\max}$  is a necessary attribute for elite endurance athletes, a stellar endurance athlete requires more than a high  $\dot{V}O_{2\max}$ , a concept discussed in [chapter 11](#).

Also, research has documented that  $\dot{V}O_{2\max}$  typically increases with physical training for only 8 to 12 weeks and then plateaus despite continued higher-intensity training. Although  $\dot{V}O_{2\max}$  does not continue to increase, participants continue to improve their endurance performance. It appears that these individuals develop the ability to perform at a higher percentage of their  $\dot{V}O_{2\max}$ . Well-trained marathon runners, for example, can complete a 42 km (26.1 mi) marathon at an average pace that equals approximately 75% to 80% of their  $\dot{V}O_{2\max}$  or higher.



**FIGURE 5.4** The relation between exercise intensity (running speed) and oxygen uptake, illustrating  $\dot{V}O_{2\max}$  in a trained and an untrained man.



Consider the case of Alberto Salazar, arguably the premier marathon runner in the world in the 1980s. His measured  $\dot{V}O_{2\max}$  was  $70 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ . That is below the  $\dot{V}O_{2\max}$  one might expect based on his best marathon performance of 2 h 8 min. He was, however, able to run at a race pace in the marathon at 86% of his  $\dot{V}O_{2\max}$ , a percentage considerably higher than that of other world-class runners. This may partly explain his world-class running ability.

Because individuals' energy requirements vary with body size,  $\dot{V}O_{2\max}$  generally is expressed relative to body weight, in milliliters of oxygen consumed per kilogram of body weight per minute ( $\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ ). This allows a more accurate comparison of the cardiorespiratory endurance capacity of different-sized individuals who exercise in weight-bearing events, such as running. In nonweight-bearing activities, such as swimming and cycling, endurance is better reflected by  $\dot{V}O_{2\max}$  measured in liters per minute.

Normally active but untrained 18- to 22-year-old college students have an average  $\dot{V}O_{2\max}$  of about  $38$  to  $42 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$  for women and  $44$  to  $50 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$  for men. In contrast, poorly conditioned adults may have values below  $20 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ . At the other end of the spectrum,  $\dot{V}O_{2\max}$  values of  $80$  to  $84 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$  have been measured for elite male long-distance runners and cross-country skiers. (The highest  $\dot{V}O_{2\max}$  value recorded for a man is that of a champion Norwegian cross-country skier who had a  $\dot{V}O_{2\max}$  of  $94 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ ! The highest value recorded for a woman is  $77 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$  for a Russian cross-country skier.)

After the age of 25 to 30 years, the  $\dot{V}O_{2\max}$  of inactive individuals decreases at a rate of about 1% per year, attributable to the combination of biological aging and sedentary lifestyle. Two physiological reasons why adult women generally have  $\dot{V}O_{2\max}$  values considerably below those of adult men (discussed further in [chapter 19](#)) are sex differences in body composition (women generally have less fat-free mass and more fat mass) and blood

hemoglobin content (lower in women, so they have a lower oxygen-carrying capacity).

### **Anaerobic Effort and Exercise Capacity**

No exercise is 100% aerobic or 100% anaerobic. The methods we have discussed thus far ignore the anaerobic processes that accompany aerobic exercise. How can the interaction of the aerobic (oxidative) processes and the anaerobic processes be evaluated? The most common methods for estimating anaerobic contribution to sustained exercise involve examination of either the excess postexercise oxygen consumption (EPOC) or the lactate threshold.

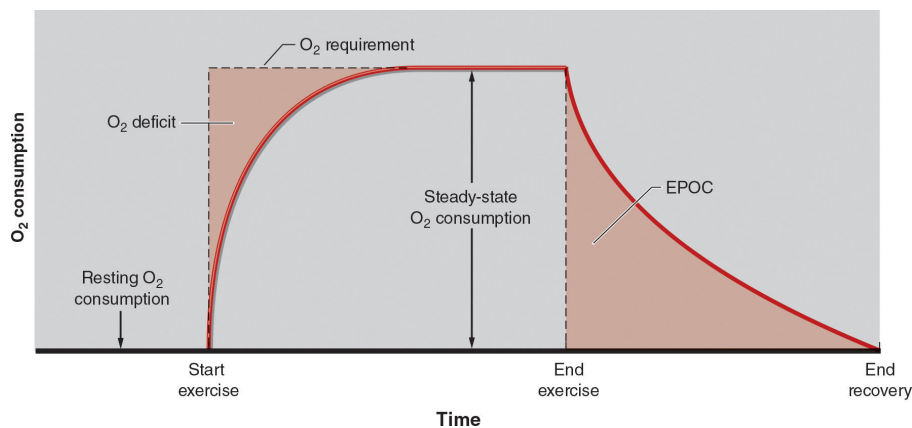
### ***Postexercise Oxygen Consumption***

The matching of energy requirements during exercise with oxygen delivery is not perfect. When aerobic exercise begins, the oxygen transport system (respiration and circulation) does not immediately supply the needed quantity of oxygen to the active muscles. Oxygen consumption requires several minutes to reach the required (steady-state) level at which the aerobic processes are fully functional, even though the body's oxygen requirements increase the moment exercise begins.

Because oxygen needs and oxygen supply differ during the transition from rest to exercise, the body incurs an **oxygen deficit**, as shown in [figure 5.5](#). This deficit accrues even at low exercise intensities. The oxygen deficit is calculated simply as the difference between the oxygen required for a given exercise intensity (steady state) and the actual oxygen consumption. Despite the insufficient oxygen delivery at the onset of exercise, the active muscles are able to generate the ATP needed through the anaerobic pathways described in [chapter 2](#).

During the initial minutes of recovery, even though active muscle activity has stopped, oxygen consumption does not immediately decrease to a resting value. Rather, oxygen consumption decreases gradually toward resting values ([figure 5.5](#)). This excess oxygen consumption, which exceeds that required at rest, was traditionally referred to as the "oxygen debt." The more common term today is **excess postexercise oxygen consumption (EPOC)**. The EPOC is the volume of oxygen consumed during the minutes immediately

after exercise ceases that is above that normally consumed at rest. Everyone has experienced this phenomenon at the end of an intense exercise bout: A fast climb up several flights of stairs leaves one with a rapid pulse and breathing hard, physiological adjustments that serve to support the EPOC. After several minutes of recovery, heart rate and breathing return to resting rates.



**FIGURE 5.5** Oxygen requirement (dashed line) and oxygen consumption (red solid line) during exercise and recovery, illustrating the oxygen deficit and the concept of excess postexercise oxygen consumption (EPOC).



For many years, the EPOC curve was described as having two distinct components: an initial fast component and a secondary slow component. According to classical theory, the fast component of the curve represented the oxygen required to rebuild the ATP and phosphocreatine (PCr) used during the initial stages of exercise. Without sufficient oxygen available, the high-energy phosphate bonds in these compounds were broken to supply the required energy. During recovery, these bonds would need to be re-formed,

via oxidative processes, to replenish the energy stores or to repay the debt. The slow component of the curve was thought to result from removal of accumulated lactate from the tissues, by either conversion to glycogen or oxidation to  $\text{CO}_2$  and  $\text{H}_2\text{O}$ , thus providing the energy needed to restore glycogen stores.

According to this theory, both the fast and slow components of the curve reflected the anaerobic activity that had occurred during exercise. The belief was that by examining the postexercise oxygen consumption, one could estimate the amount of anaerobic activity that had occurred.

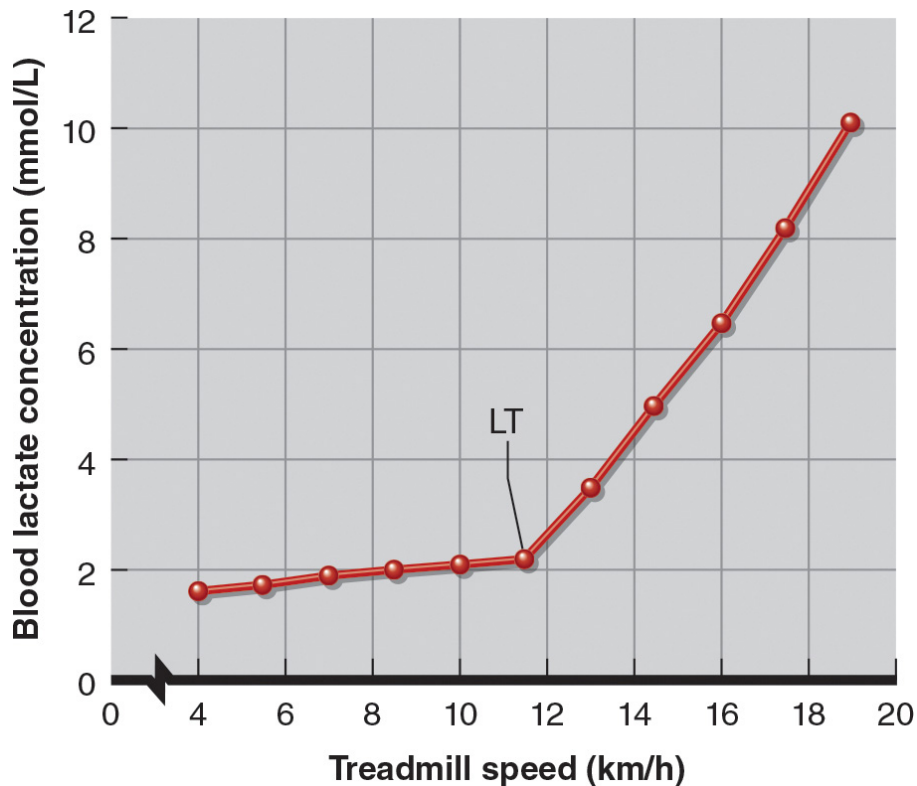
However, more recently researchers have concluded that the classical explanation of EPOC is too simplistic. For example, during the initial phase of exercise, some oxygen is borrowed from the oxygen stores (hemoglobin and myoglobin), and that oxygen must be replenished during early recovery as well. Also, respiration remains temporarily elevated following exercise partly in an effort to clear  $\text{CO}_2$  that has accumulated in the tissues as a by-product of metabolism. Body temperature also is elevated, which keeps the metabolic and respiratory rates high, thus requiring more oxygen; and elevated concentrations of norepinephrine and epinephrine during exercise have similar effects.

Thus, the EPOC depends on many factors other than merely the replenishing of ATP and PCr and the clearing of lactate produced by anaerobic metabolism.

### ***Lactate Threshold***

Many investigators consider the lactate threshold a good indicator of an athlete's potential for endurance exercise. The **lactate threshold** is defined as the point at which blood lactate begins to substantially accumulate above resting concentrations during exercise of increasing intensity. For example, a runner might be required to run on the treadmill at different speeds with a rest between each speed. After each run, a blood sample is taken and blood lactate is measured. [Figure 5.6](#) depicts the relation between blood lactate and running velocity. At low running velocities, blood lactate concentrations remain near resting levels. But as running speed increases, the blood lactate concentration increases rapidly beyond

some threshold exercise intensity. The point at which blood lactate first appears to increase disproportionately above resting values is called the lactate threshold.



**FIGURE 5.6** The relation between exercise intensity (running velocity) and blood lactate concentration. Blood samples were taken from a runner's arm vein and analyzed for lactate after the subject ran at each speed for 5 min. LT = lactate threshold.

The lactate threshold has been thought to reflect the interaction of the aerobic and anaerobic energy systems. Some researchers have suggested that the lactate threshold represents a significant shift toward anaerobic glycolysis, which forms lactate from pyruvic acid. Consequently, the sudden increase in blood lactate with increasing effort has also been referred to as the *anaerobic threshold*. However, blood lactate concentration is determined not only by the production of lactate in skeletal muscle or other tissues but also by the clearance of lactate from the blood by the liver and its use as a fuel source by muscle and other tissues in the body. Thus, lactate threshold is best defined as that point in time during exercise of increasing intensity when the rate of lactate production exceeds the rate of lactate clearance.

The lactate threshold is usually expressed as the percentage of maximal oxygen uptake ( $\% \dot{V}O_{2\max}$ ) at which it occurs. In untrained people, the lactate threshold typically occurs at approximately 50% to 60% of their  $\dot{V}O_{2\max}$ , while elite endurance athletes may not reach lactate threshold until closer to 70% or 80% of  $\dot{V}O_{2\max}$ .

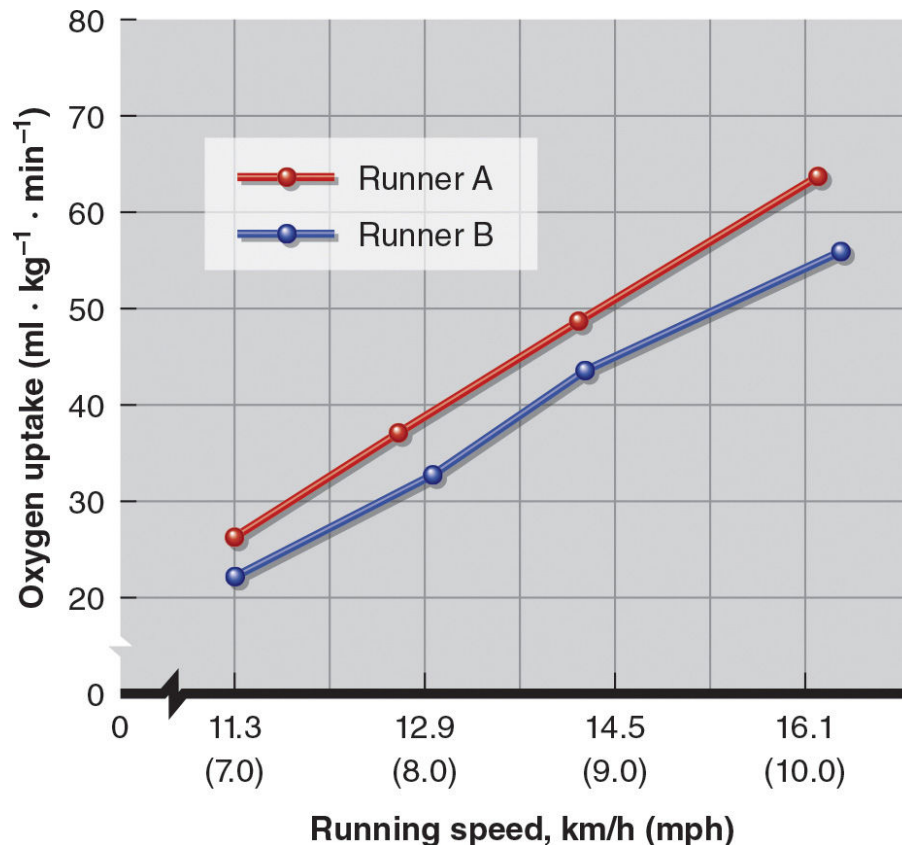
From the previous section, we learned that in addition to a high  $\dot{V}O_{2\max}$ , the percentage of  $\dot{V}O_{2\max}$  that an athlete can maintain for a prolonged period is a major determinant of successful endurance performance. The lactate threshold is likely the major determinant of the fastest pace that can be tolerated during a long-term endurance event. So the ability to perform at a higher percentage of  $\dot{V}O_{2\max}$  probably reflects a higher lactate threshold. Consequently, a lactate threshold at 80%  $\dot{V}O_{2\max}$  suggests a greater aerobic exercise tolerance than a threshold at 60%  $\dot{V}O_{2\max}$ . Generally, in two individuals with the same maximal oxygen uptake, the person with the highest lactate threshold usually exhibits the best endurance performance, although other factors contribute as well, including economy of movement.

### **Economy of Effort**

As people become more skilled at performing an exercise, the energy demands during exercise at a given pace are reduced. In a sense, people become more economical. (Note that we avoid calling this efficiency, which has a more stringent mechanical definition.) This is illustrated in [figure 5.7](#) by the data from two distance runners. At all running speeds faster than 11.3 km/h (7.0 mph), runner B used significantly less oxygen than runner A. These men had similar  $\dot{V}O_{2\max}$  values (64-65 ml  $\cdot$  kg<sup>-1</sup>  $\cdot$  min<sup>-1</sup>), so runner B's lower submaximal energy use would be a decided advantage during competition.

These two runners competed on numerous occasions. During marathon races, they ran at paces requiring them to use 85% of their  $\dot{V}O_{2\max}$ . On average, runner B beat runner A by 13 min in their competitions. Because their  $\dot{V}O_{2\max}$  values were so similar but their energy needs so different during these events, much of runner B's competitive advantage could be attributed to his greater running

economy. Unfortunately, there is no single specific explanation for the underlying causes of differences in economy, which are likely due to a variety of complex physiological and biomechanical factors.



**FIGURE 5.7** The oxygen requirements for two distance runners running at various speeds. Although they had similar  $\dot{V}O_{2\max}$  values ( $64\text{--}65 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ ), runner B was more economical and therefore could run at a faster pace for a given oxygen cost.



Various studies with sprint, middle-distance, and distance runners have shown that marathon runners are generally very economical. It is not uncommon for ultra-long-distance runners to use 5% to 10% less energy than middle-distance runners and sprinters at a given pace. However, this economy of effort has been studied at only relatively slow speeds (paces of 10–19 km/h, or 6–12 mph). We can

reasonably assume that distance runners are less economical at sprinting than runners who train specifically for short, faster races. It is probable that runners self-select their chosen events in part because they achieve early success, success achieved in part due to better running economy at that distance.

Variations in running form and the specificity of training for sprint and distance running may account for at least part of these differences in running economy. Film analyses reveal that middle-distance runners and sprinters have significantly more vertical body movement when running at 11 to 19 km/h (7-12 mph) than marathoners do. But such speeds are well below those required during middle-distance races and probably do not accurately reflect the running economy of competitors in shorter events of 1,500 m (1 mi) or less.

Performance in other athletic events might be even more affected by economy of movement. Part of the energy expended during swimming, for example, is used to support the body on the surface of the water and to generate enough force to overcome the water's resistance to motion. Although the energy needed for swimming depends on body size and buoyancy, the efficient application of force against the water is the major determinant of swimming economy.

### **Characteristics of Successful Athletes in Aerobic Endurance Events**

From our discussion of the metabolic characteristics of aerobic endurance athletes in this chapter and of their muscle fiber type characteristics in [chapter 1](#), it is clear that to be successful in aerobic endurance activities, one needs some combination of the following:

- High  $\dot{V}O_{2\max}$
- High lactate threshold when expressed as a percentage of  $\dot{V}O_{2\max}$
- High economy of effort, or a low  $\dot{V}O_2$  for a given absolute exercise intensity
- High percentage of type I muscle fibers

From the limited data available, these four characteristics appear to be properly ranked in their order of importance. As an example,

running velocity at lactate threshold and  $\dot{V}O_{2\max}$  are the best predictors of actual race pace among a group of elite distance runners. However, each of those runners already has a high  $\dot{V}O_{2\max}$ , which, in elite athletes, is supported by having a large heart and an expanded blood volume. Although economy of effort is important, it does not vary much between elite runners. Finally, having a high percentage of type I muscle fibers is helpful but not essential. The bronze medal winner in one of the Olympic marathon races had only 50% type I muscle fibers in his gastrocnemius muscle, one of the primary muscles used in running.

### **Energy Cost of Various Activities**

The amount of energy expended for different activities varies with the intensity and type of exercise. Despite subtle differences in economy, the *average* energy costs of many activities have been determined, usually through the measurement of oxygen consumption during the activity to determine an average oxygen uptake per unit of time. The amount of energy expended per minute (kcal/min) then can be calculated from this value.

These values typically ignore the anaerobic aspects of exercise and the EPOC. This omission is important because an activity that costs a total of 300 kcal during the actual exercise period may cost an additional 100 kcal during the recovery period. Thus, the total cost of that activity would be 400, not 300, kcal. Because of these nuances along with individual variation, the “calories burned” readouts on exercise machines can be highly inaccurate.

The body requires 0.16 to 0.35 L of oxygen per minute to satisfy its resting energy requirements. This would amount to 0.80 to 1.75 kcal/min, 48 to 105 kcal/h, or 1,152 to 2,520 kcal/day. Obviously, any activity above resting levels will add to the projected daily expenditure. The range for total daily caloric expenditure is highly variable and depends on

- physical activity (by far the largest influence),
- age,
- sex,
- body size,

- weight, and
- body composition.

The energy costs of sport activities also differ. Some, such as archery or bowling, require only slightly more energy than rest. Others, such as sprinting, require such a high rate of energy delivery that they can be maintained for only seconds. Clearly, both exercise intensity and duration of the activity must be considered to determine energy expended. For example, approximately 29 kcal/min is expended during running at 25 km/h (15.5 mph), but this pace can be endured for only brief periods. Jogging at 11 km/h (7 mph), on the other hand, expends only 14.5 kcal/min, half that of running at 25 km/h (15.5 mph). But jogging can be maintained for considerably longer, resulting in greater total energy expenditure for an exercise session.

Table 5.2 provides estimates of average energy expenditure during various activities for men and women. Remember that these values are merely averages, and these figures vary considerably with individual differences such as those on the preceding list and with individual skill (economy of movement).

## In Review

- The basal metabolic rate (BMR) is the minimum amount of energy required by the body to sustain basic cellular functions and is related to fat-free body mass and, to a lesser extent, body surface area. It typically ranges from 1,100 to 2,500 kcal/day, but when daily activity is added, typical daily caloric expenditure is 1,700 to 3,100 kcal/day.
- $\dot{V}O_2$  increases linearly with increased exercise intensity but eventually reaches a plateau. Its maximal value is called the  $\dot{V}O_{2max}$ . When volitional fatigue limits exercise before a true maximum is reached, the term  $\dot{V}O_{2peak}$  is used.
- Successful aerobic performance is linked to a high  $\dot{V}O_{2max}$ , to the ability to perform for long periods at a high percentage of  $\dot{V}O_{2max}$ , to the running velocity at lactate threshold, and to a good economy of movement.
- The EPOC is the elevated metabolic rate above resting levels that occurs during the recovery period immediately after exercise has ceased.
- Lactate threshold is that point at which blood lactate production begins to exceed the body's ability to clear lactate, resulting in a rapid increase in blood lactate